

Candidate Number

Candidate Name _____

INTERNATIONAL ENGLISH LANGUAGE TESTING SYSTEM

Academic Reading

Additional materials:

Answer sheet for Listening and Reading

Time 1 hour

INSTRUCTIONS TO CANDIDATES

Do not open this question paper until you are told to do so.
Write your name and candidate number in the spaces at the top of this page. Read the instructions for each part of the paper carefully.

Answer all the questions.

Write your answers on the answer sheet. Use a pencil.

You must complete the answer sheet within the time limit.

At the end of the test, hand in both this question paper and your answer sheet.

INFORMATION FOR CANDIDATES

There are **40** questions on this question paper.

Each question carries one mark.

READING PASSAGE 1

*You should spend about 20 minutes on **Questions 1-13**, which are based on Reading Passage 1 on pages 2 and 3.*

Andrew Carnegie: industrialist and philanthropist

The Scottish-born American industrialist and philanthropist Andrew Carnegie, who was the leader of the American steel industry from 1873 to 1901, donated large sums of his fortune to educational, cultural, and scientific institutions. He was born in 1835 in Scotland, the son of William Carnegie, a weaver, and Margaret Morrison Carnegie.

With the invention of weaving machines, William Carnegie's job disappeared, and eventually the family was forced into poverty. In 1848 the family left Scotland and settled in Pittsburgh, Pennsylvania, in the USA. William found a job in a cotton factory, but he soon quit to return to his home handloom, making linens and trying to sell them door-to-door. Andrew Carnegie also worked in the cotton factory, but after his father died in 1855, his strong desire to help take care of the family pushed him to educate himself. He became an avid reader, a theatregoer, and a lover of music.

Andrew Carnegie became a messenger boy in a telegraph office and later became a telegraph operator. Thomas A. Scott, superintendent of the western division of the Pennsylvania Railroad, made the eighteen-year-old his secretary. Carnegie was soon earning enough to buy a house for his mother. During the Civil War in the United States (1861-65), he helped organize the military telegraph system, but soon returned to Pittsburgh to take Scott's old job with the railroad.

Between 1865 and 1870 Carnegie made money through investments in several small iron mills and factories. He also traveled throughout England, selling the bonds of small United States railroads and bridge companies. Carnegie began to see that steel was eventually going to replace iron for the manufacture of rails, structural shapes, pipe, and wire. In 1873 he organized a steel rail company and continued building by cutting prices, driving out competitors, shaking off weak partners, and putting earnings back into the company. He never went public (sold shares of his company in order to raise money). Instead he obtained capital from profit, and when necessary from banks, and he kept on growing, making heavy steel alone.

In the 1880s Carnegie's purchases included a majority stake in the H. C. Frick Company in Pennsylvania that had vast coal interests, and also the Homestead mills business. Frick became his partner and eventually chairman of the Carnegie Company. Carnegie had moved to New York City in 1867 to be close to the marketing centers for steel products, while Frick stayed in Pennsylvania as the general manager. They made a good team.

Behind the scenes, Carnegie planned new projects and the improvement of existing plants; Frick was the working director who watched over the mass-production programs that helped keep prices down.

Carnegie spent his leisure time traveling. He also wrote several books, including *Triumphant Democracy* (1886), which pointed out what he thought were the advantages of life in the United States over the unequal societies of Britain and other European countries. To Carnegie, access to education was the key to America's political stability and industrial accomplishments. In 1889 he published an article, 'Wealth', stating his belief that the rich had a duty to use their money to improve the welfare of the community. Carnegie remained a bachelor until his mother died in 1886. A year later he married Louise Whitfield and they had one child together. The couple began to spend six months each year in Scotland, though Carnegie kept an eye on business developments and problems.

Carnegie's absence from the United States was a factor in the Homestead mills strike of 1892. After acquiring Homestead, Carnegie had invested in new plants and equipment, increased production, and automated many of the mill's operations, cutting down the number of workers that were needed. Carnegie believed that workers had a right to join unions and to bargain with management through their unions. He also recognized the right to strike, as long as the action was conducted peacefully. He viewed strikes as trials of strength, with peaceful discussion resolving the conflict. In contract talks during 1892, Frick wanted to lower the minimum wage because of the need for fewer workers. The union representing his workers would not accept this and organized a strike.

Following a violent confrontation that resulted in the death of eight workers, soldiers were sent in to restore order and Carnegie was criticized for his lack of action. In the 1890s Carnegie also began to meet with tougher competition from newer, bigger companies who were interested in controlled prices and sharing the market. Companies that he had sold to for years threatened to cut down their purchases unless he agreed to cooperate. These threats made him decide to fight back. He refused to enter into any agreements with other companies. Moreover, he decided to invade their territories by making similar products and by expanding his sales activities into the west of the country. Eventually, though, he decided to sell his company to the newly formed U.S. Steel Corporation in 1901 for almost \$500 million. Carnegie's personal share was \$225 million.

In retirement, Carnegie began to set up trust funds 'for the improvement of mankind'. He built some three thousand public libraries all over the English-speaking world and funded galleries, museums and concert halls in the US. He also built a group of technical schools, a research institute, and a foundation providing pensions for university professors. When Carnegie died in 1919, the continuation of his broad interests was put under the general charge of the Carnegie Corporation, with a donation of \$125 million.

Questions 1-7

Complete the notes below.

*Choose **ONE WORD ONLY** from the passage for each answer.*

Write your answers in boxes 1-7 on your answer sheet.

Carnegie's Life

1835-1855

- found employment in the same workplace as his 1
- educated himself through books, theatrical performances and 2

1855-1865

- worked in a telegraph office
- became 3 to Thomas A. Scott at Pennsylvania Railroad

1865-1870

- made investments in the US and England
- realised that changing engineering needs would require a greater demand for 4

1873

- managed a rail manufacturing business by:
 - reducing 5
 - restructuring
 - reinvesting profit and sometimes borrowing money from 6

1880s

- Frick and Carnegie became partners
- Carnegie's responsibilities: arranging future 7 and innovations for factories
- Frick's responsibilities: production

Questions 8-13

Do the following statements agree with the information given in Reading

Passage 1? In boxes 8-13 on your answer sheet, write

TRUE *if the statement agrees with the information*

FALSE *if the statement contradicts the information*

NOT GIVEN *if there is no information on this*

8. Carnegie believed that the USA offered more opportunities than European countries did.

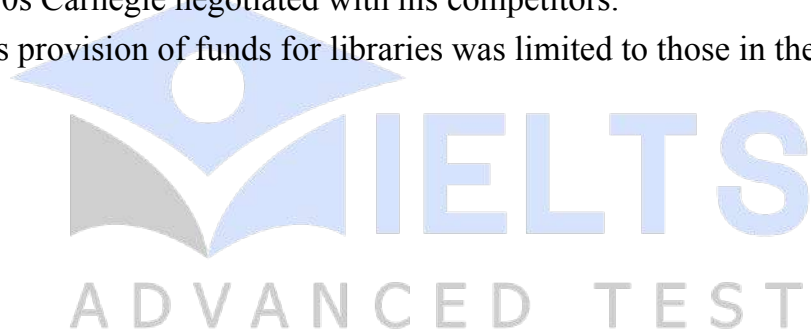
9. A smaller workforce was required at the Homestead factory after Carnegie took control.

10. Carnegie prevented his workers from joining unions.

11. Carnegie blamed Frick for the violent workers' strike in 1892.

12. In the 1890s Carnegie negotiated with his competitors.

13. Carnegie's provision of funds for libraries was limited to those in the United States.



READING PASSAGE 2

You should spend about 20 minutes on Questions 14–26, which are based on Reading Passage 2 below.

Questions 14 – 20

Reading Passage 2 has seven paragraphs, A–G.

Choose the correct heading for each paragraph from the list of headings below.

Write the correct number, i–xi, in boxes 14–20 on your answer sheet

List of Headings

- i. Training coin makers
- ii. A form of regional promotion
- iii. More gold than silver in Greece
- iv. Calculating what a coin was worth
- v. A simpler, more efficient way of trading
- vi. Putting a face to famous historical names
- vii. An uncertain future
- viii. Careless workmanship caused breakages
- ix. Unchanging coins signalled trustworthiness
- x. Reading the writing on coins
- xi. Significant records of past societies

14 Paragraph A

15 Paragraph B

16 Paragraph C

17 Paragraph D

18 Paragraph E

19 Paragraph F

20 Paragraph G

Coins – the first form of money

A Few people pay much attention to coins; we take them for granted. Since the advent of paper currency and the abandonment of a monetary system based on intrinsic of actual value, they have been demoted to the status of change: what is left when you have paid for something with 'paper' money. There is even a belief that they will one day be obsolete, as will paper or plastic notes, and money will at last have achieved the pure status of a mathematical formulation, a system of electronic credits and debits.

B In the modern media environment, the sheer number of visual images is overwhelming, but it makes one stop to think that coins are among the most durable of our artefacts, and future archaeologists will scrutinise coins in order to learn about their distant ancestors. This is exactly the way ancient coins have informed us for over four millennia. There are thousands of different coins, each designed to carry meaning as well as value, tiny but rich sources of documentary evidence about events and cultural attitudes, frequently produced with exquisite craftsmanship.

C Coins in Europe date from 2,500 to 2,700 years ago, and seem to have originated in the ancient kingdom of Lydia, in what is now Turkey. Before coins, trade was based on barter: the direct trade of commodities. Barter required the physical presence of both commodities that were being traded. Metal coins were far more portable, and not subject to the same spoilage on long sea journeys as a commodity like grain might be if it became wet. However, they could also be accumulated, and these hoards could be stolen. Coins were therefore a temptation to criminals. The Greeks, however, were the first to realise the full potential of the new invention. They were a trading people of seafarers. Their civilisation, which was composed of dozens of city-states, was highly decentralised. Despite the potential for theft, the Greeks realised that they could benefit enormously from a system that used coins, because it meant that they no longer had to carry goods with them to barter.

D The value of coins was generally based on the intrinsic value of the metal they were made of. At first this was mostly silver, and this meant that the discovery of silver mines on Greek territory actually increased the quantity of money they could produce. Had the Greek mining been on a larger scale, it could have caused problems similar to those faced by the Spanish centuries later, when so much gold from the New World was brought into their economy that it resulted in serious inflation. One advantage of intrinsic value was that coins could be weighed to determine the amount of metal they contained, so that the

only thing an issuing state had to guarantee was the purity of the metal to prevent accusations of fraud and cheating, which could cause fighting and unrest. This was quite difficult to do, especially with coins made of gold, until Archimedes realised that different metals have different weight-volume ratios, and that by measuring the amount of displacement when coins were immersed in water, it was possible to establish whether gold was alloyed with silver.

E All ancient coins were made by hand using engraved metal moulds. Many hundreds of coins were made, and as a result the moulds often cracked and new ones had to be engraved. This explains the endless variety encountered in coins with the same designs because the moulds had to be remade repeatedly, by many different artisans. One of the distinctive features of coins, in fact, is that they are the work of anonymous craftsmen, and yet each coin is an intimate reflection of the character of its city of origin. It was the Greeks who realised that coins could have a double function, as a unit of value and as a symbol of the identity of the city that issued them. The Greek view of life was deeply motivated by a spirit of competition, and coins, which circulated around the trading world, were a way of advertising the qualities of each particular city-state.

F Interestingly, Greek coins from the city of Athens, which was the home of the greatest sculptors of the classical period, seem to have been completely disconnected from the high sculptural tradition. The beautiful features of the city's patron, Athena (with her owl on the reverse of the coin), seem to have been repeated virtually unaltered for many years. Perhaps this was because the Athenian currency had grown increasingly to be the international standard and, like the US dollar today, a uniform design implied integrity.

G One side of a coin was always occupied by the head of a divinity, so it was a relatively small step, at around 323 BC, for Greek coins to replace that head with the portrait of the reigning monarch, a tradition that has continued in monarchies to the present day. Republics may have either symbolic heads going back to the older tradition of divinities or the features of a distinguished past head of state. The Romans adopted the Greek practice of coin design and as a result, the features of many great Romans have survived and are known to modern generations, despite the fact that the art of portrait painting was completely lost for more than a millennium following the fall of the Roman Empire.

Questions 21 and 22

Choose **TWO** letters, **A-E**.

Write the correct letters in boxes 21 and 22 on your answer sheet.

Which **TWO** of these comments about ancient coins are made by the writer of the text?

- A. There is no record of the names of the coin makers.
- B. Coins were not popular as payment for traded objects.
- C. Coins with the same design often looked quite different.
- D. Greek coins were more beautiful in design than Roman coins.
- E. At times, too many coins were produced in Greece.

Questions 23 – 26

Complete the summary below.

Choose **ONE WORD ONLY** from the passage for each answer.

Write your answers in boxes **23-26** on your answer sheet.

Greek traders and coins

The ancient Greek **23** was made up of many city-states which were spread over a large area. Many Greeks were seafarers who travelled the seas trading. Before coins, they had to do this through **24** This meant that, whenever they set out, they had to carry a lot of goods, which could get damaged on the voyages. The Greeks liked coins; they were made of metal, and the weather and sea had little effect on them. The Greeks had a plentiful supply of **25** which they mined themselves, but were able to avoid the problem of **26** which the Spanish encountered much later, after they allowed too much gold from the New World to enter their economy.

READING PASSAGE 3

You should spend about 20 minutes on Questions 27–40, which are based on Reading Passage 3 on pages 10 and 11.

Music soothes and awes – and may help us heal

Tapping into the brain's ability to process music, researchers are using songs and sounds in novel ways

Dan Ellsey, 33, was sitting in his wheelchair, his virtually useless arms and weak torso strapped to the chair for safety. As we were introduced, he arched his back, grinned broadly, and aimed the power of his dark brown eyes at me, as if eye contact were his only means of transcending the prison of his body.

But it isn't. In the last few years, Ellsey, who was born with cerebral palsy, has discovered another, almost miraculous way of expressing himself: composing music with a special computerized system called Hyperscore. For Ellsey, as for most human beings, music has an almost inexplicable power. It can rouse armies to battle, soothe babies to sleep, and communicate peaks of joy and depths of sorrow that mere words cannot.

Why evolution would have endowed our brains with the neural machinery to make music is a mystery. 'It's unclear why humans are so uniquely sensitive to music. Certainly, music shares many features with spoken language, and our brains are particularly developed to process the rapid tones and segments of sound that are common to both,' said neurologist and author Dr Oliver Sacks, whose latest book is *Musicophilia: Tales of Music and the Brain*. 'Some researchers,' he added in an e-mail interview, 'think that in primitive cultures, music and speech were not distinct. Other researchers debate which came first in evolution, speech or song.' What is clear is that the brain is abundantly wired to process music.

Scientists at the Montreal Neurological Institute and Hospital, for instance, have found dramatic evidence on brain scans that the chills, or visceral feelings of awe, that people report when listening to their favourite music are real. Reactions to music appear to vary greatly. Music that a person likes, but not music that is disliked, activates the higher thinking centers in the brain's cortex and, perhaps more importantly, the 'ancient circuitry, the motivation and reward system,' according to experimental psychologist Robert Zatorre, a member of the Montreal team. This primeval part of the brain also governs basic drives such as hunger, thirst and sex, suggesting that the brain may consider music on a par with them.

Music has the power not just to awe but also to heal. If a person has a stroke on the left side of the brain, where the speech centers in most people are, that 'wipes out a major part of communication,' said Dr Gottfried Schlaug, chief of the Cerebrovascular Disorder

Division and Stroke Recovery Laboratory at Beth Israel Deaconess Medical Center. But if the right side, where a lot of music is processed, is intact, some stroke patients can use melodic intonation therapy, which involves singing using two tones relatively close in pitch, to communicate. Schlaug's research suggests that with intensive therapy, some patients can even move from this two-tone singing back to actual speech.

Stroke patients with gait problems also profit from music therapy. At the Center for Biomedical Research in Music at Colorado State University in Fort Collins, director Michael Thaut and his team have shown that people who are partially paralyzed on one side of the body can be retrained to walk faster and in a more coordinated way if they practice walking rhythmically, cued by music or a metronome. Combining rhythmic training with physical therapy also helps stroke patients recover gait faster. 'Music helps us organize our movement,' said Kathleen Howland, a music therapist who teaches at Lesley University in Cambridge in the USA. In fact, researchers have found that auditory signals are more powerful than visual signals in coordinating movement in healthy people.

Patients undergoing colonoscopy seem to feel less pain and need fewer sedative drugs if they listen to music during the procedure, according to several studies. But not all studies have been so clear-cut. A 2007 review by the Cochrane Collaboration, a nonprofit, international organization that evaluates medical research, pooled data from 51 pain studies and concluded that, although listening to music can reduce the intensity of pain and the need for narcotic drugs, the benefit, overall, was small.

A number of studies, however, show that music therapy can reduce pain, such as a 2001 study on burn patients, whose burns must be frequently scraped to reduce dead tissue. Music therapy may improve mental state and functioning in people with schizophrenia, and communication in children with autistic spectrum disorders, according to Cochrane reviews. Babies, as any parent knows, clearly respond to music. Premature infants who listen to lullabies learn to suck better and gain more weight than those who don't get music therapy. Deforia Lane, director of music therapy at the University Hospitals Ireland Cancer Center in Cleveland, has found an improvement in immune response among hospitalized children who played, sang and created music, compared with children who did not get music therapy. Indeed, the list of potential benefits from music therapy seems long. And for people such as Ellsey, the benefits can be nothing short of liberating.

Questions 27–32

Do the following statements agree with the views of the writer in Reading Passage 3? In boxes 27–32 on your answer sheet, write

YES *if the statement agrees with the views of the writer*

NO *if the statement contradicts the views of the writer*

NOT GIVEN *if it is impossible to say what the writer thinks about this*

27 Dan Ellsey has been able to communicate through music since he was very young.

28 Armies respond to music that is loud and rhythmic.

29 Researchers have determined why people react so emotionally to music.

30 Both music and speech are composed of sound segments.

31 It is clear that song developed more quickly than speech.

32 Emotional reactions to music have been scientifically demonstrated.

Questions 33–35

Choose the correct letter, A, B, C or D.

Write the correct letter in boxes 33–35 on your answer sheet.

33. According to the article, music stimulates higher brain activity when it

- A.** provokes hunger.
- B.** requires thought.
- C.** is soothing.
- D.** is pleasing.

34 According to Dr Schlaug's research, some stroke patients may regain speech if

- A.** they practise making sounds that are close together.
- B.** they reactivate the left side of the brain.
- C.** they receive music therapy daily.
- D.** they listen to tunes in groups.

35 According to the Cochrane Collaboration review, some research into the link between music and pain relief suggests

- A.** the value of music is underestimated.
- B.** the value of music may be limited.
- C.** more studies are needed in these areas.
- D.** patients welcome this research.

Questions 36–40

Complete each sentence with the correct ending, A–H, below.

Write the correct letter, A–H, in boxes 36–40 on your answer sheet.

List of Endings

A rely less on drugs.

B get heavier.

C exhibit improved social interaction.

D experience psychological benefits.

E sleep more soundly.

F suffer less discomfort.

G accept their situation more easily.

H have stronger resistance to disease.

36 Burn victims who receive music therapy

37 People with schizophrenia who receive music therapy

38 Autistic children who receive music therapy

39 Premature babies who receive music therapy

40 Hospitalised children who receive music therapy